

Ex.NO:	DATA VISUALIZATION USING MATPLOTLIB
DATE:	

Aim

To create and analyze various data visualizations such as Line Charts, Bar Charts, Pie Charts, Histograms, Scatter Plots, and Box Plots using Matplotlib for effective data interpretation.

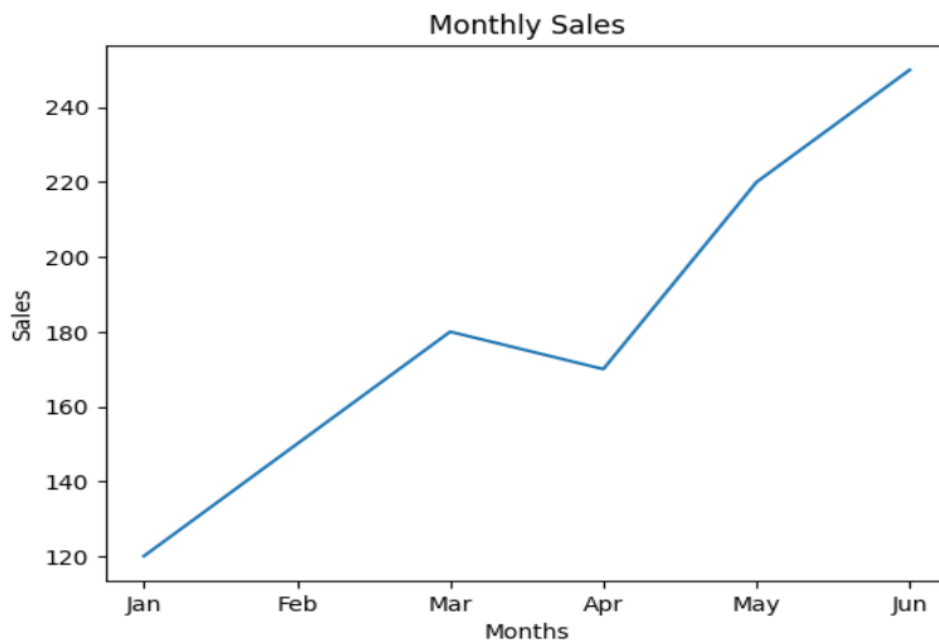
Algorithm

Code and Output

```
import matplotlib.pyplot as plt  
months = ["Jan","Feb","Mar","Apr","May","Jun"]  
sales = [120,150,180,170,220,250]  
plt.plot(months,sales)  
plt.title("Monthly Sales")  
plt.xlabel("Months")
```

```
plt.ylabel("Sales")
```

```
plt.show()
```



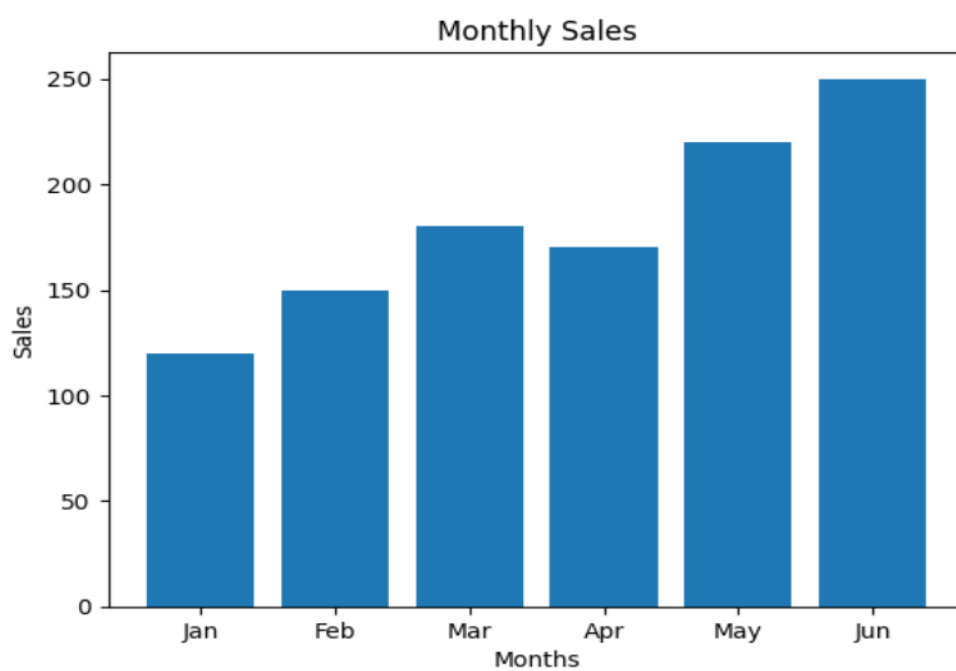
```
plt.bar(months,sales)
```

```
plt.title("Monthly Sales")
```

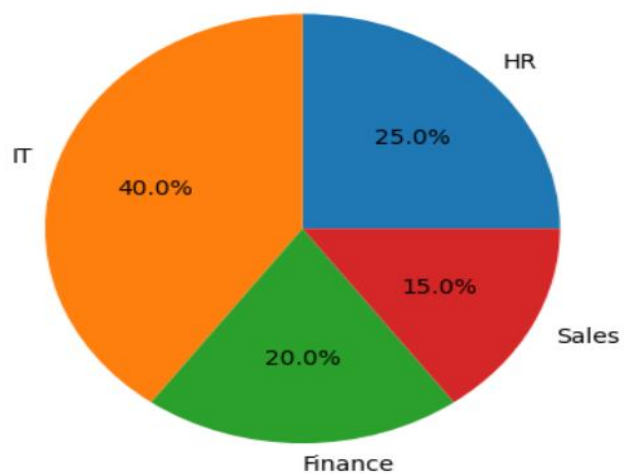
```
plt.xlabel("Months")
```

```
plt.ylabel("Sales")
```

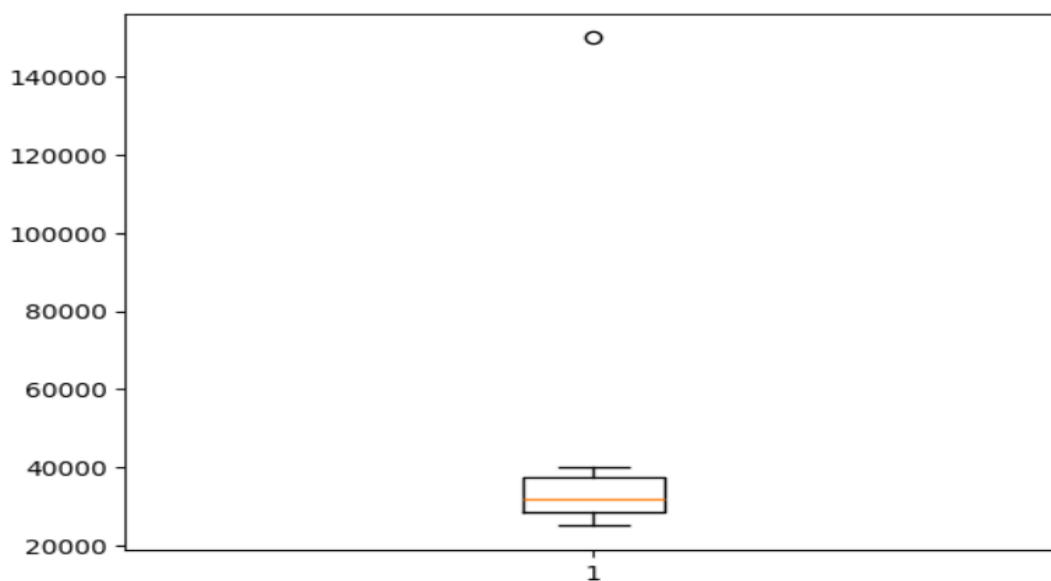
```
plt.show()
```



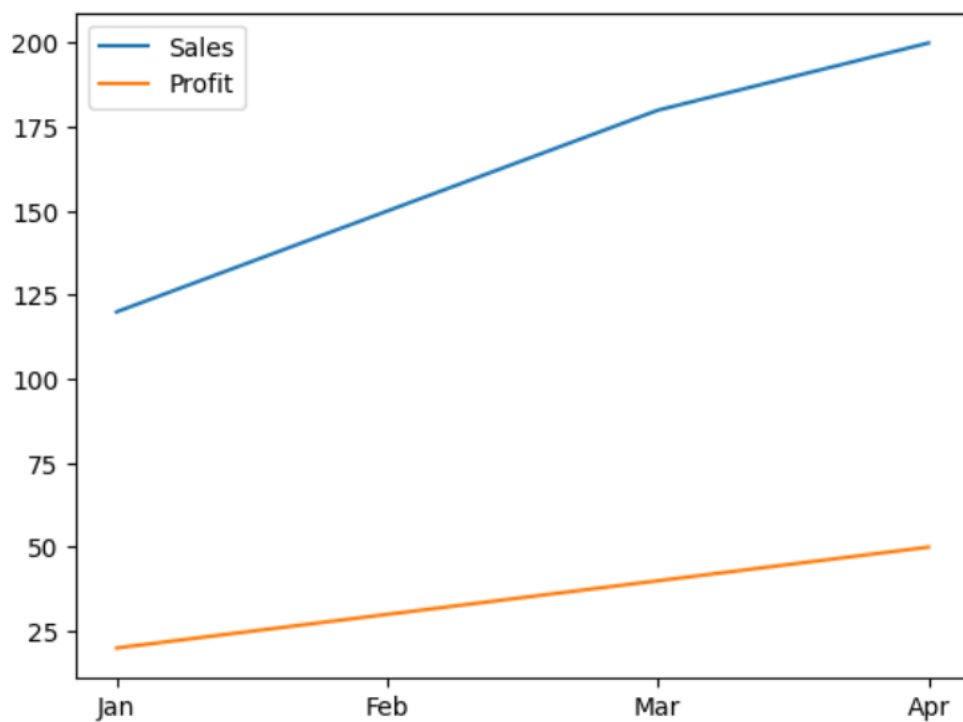
```
departments = ["HR","IT","Finance","Sales"]
employees = [25,40,20,15]
plt.pie(employees,
labels=departments,
autopct='%1.1f%%')
plt.show()
```



```
salary=[25000,27000,30000,
32000,35000,
40000,150000]
plt.boxplot(salary)
plt.show()
```



```
months=["Jan","Feb","Mar","Apr"]  
sales=[120,150,180,200]  
profit=[20,30,40,50]  
plt.plot(months,sales,label="Sales")  
plt.plot(months,profit,label="Profit")  
plt.legend()  
plt.show()
```



Result

Thus, various visualization techniques such as Line Charts, Bar Charts, Pie Charts, Histograms, Scatter Plots, and Box Plots were successfully created using Matplotlib. The graphical representations helped in understanding trends, distributions, comparisons, and relationships within the dataset.